

Data Analytics and Mining

Correlation, Hypothesis testing, Chi-square test

Data Analytics and Mining, 2026

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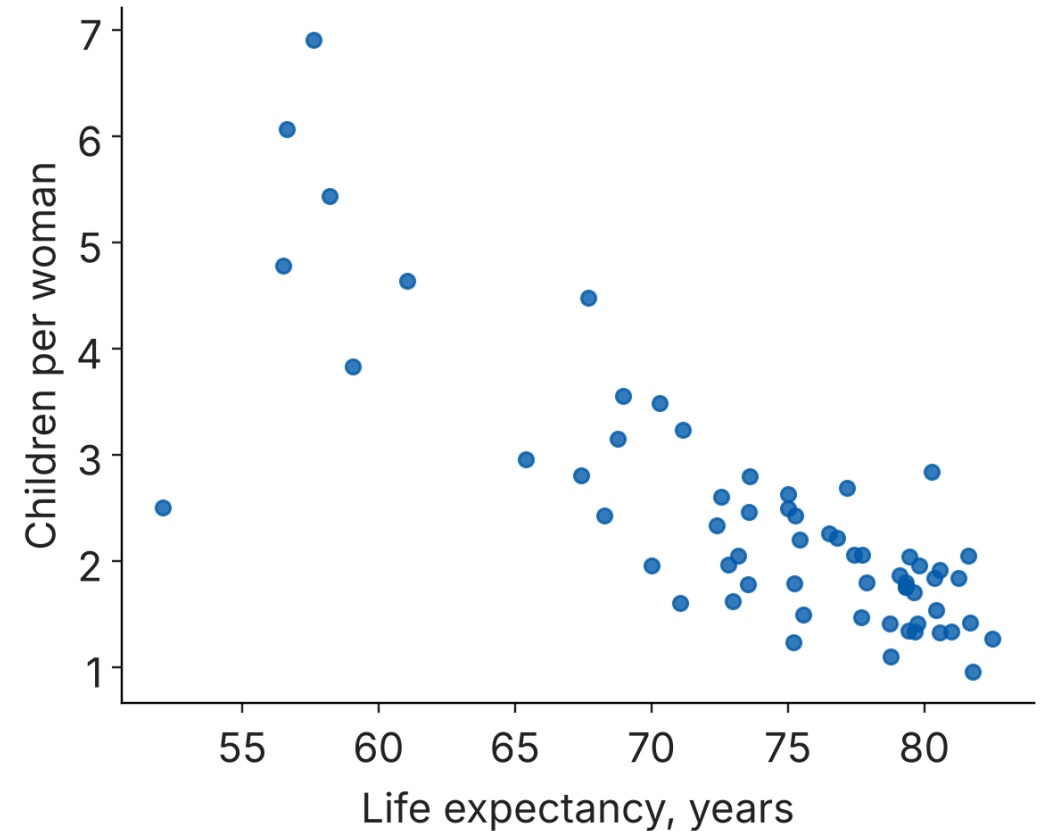
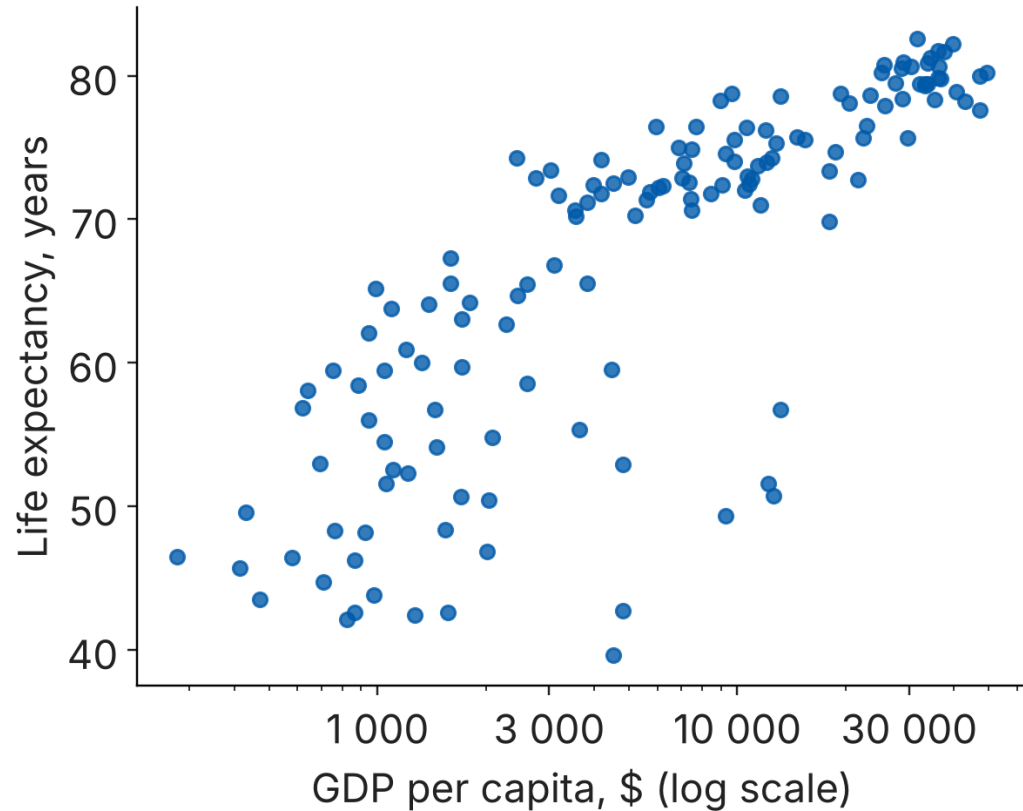


October 09, 2026

Correlation



How does a relationship look in data?



(b) Life expectancy and fertility, 62 countries, 2005. Where people live longer, women tend to have fewer children: a **negative** relationship.

What is correlation?

Correlation is a statistical technique that is used to measure and describe the relationship between two numerical variables.

Correlation is a **descriptive statistic**. Just as a sample mean gives a concise description of an entire sample, a correlation gives a concise description of a relationship.

A correlation requires **two scores for each individual** (one score from each of the two variables). These scores are usually called X and Y. The pairs of scores can be listed in a table, or shown as points in a **scatter plot**: one point per individual.

Data to compute a correlation:

family income (X) and average grade (Y) of $n = 14$ high-school students, listed from the lowest to the highest income.

Person	Family Income (in \$1000)	Student's Average Grade
A	31	72
B	38	86
C	42	81
D	44	78
E	49	85
F	56	80
G	58	91
H	65	89
I	70	94
J	90	83
K	92	90
L	106	97
M	135	89
N	174	95

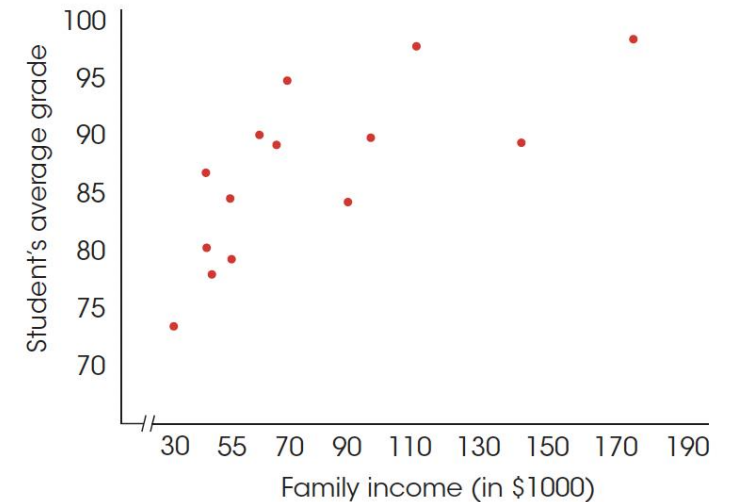


Figure: Gravetter & Wallnau, Statistics for the Behavioral Sciences

Characteristics of a relationship

The direction of the relationship

The sign of the correlation, positive or negative, describes the direction.

Positive correlation: X and Y tend to change in the same direction. When X increases from one individual to another, Y also tends to increase (income and life expectancy).

Negative correlation: X and Y tend to go in opposite directions. When X increases, Y tends to decrease (life expectancy and number of children).

The form of the relationship

The most common correlation (Pearson's r) measures **straight-line (linear)** relationships. Other forms exist, for example curved ones, and r can miss them completely.

The strength (consistency) of the relationship

How closely do the points follow a line? In a perfect linear relationship, every time X increases by one point, Y changes by exactly the same amount. Real relationships are usually not perfect.

Strength of a relationship

The strength is measured by the numerical value of the correlation, which is always between -1 and $+1$.

- **$r = +1$ or $r = -1$** : a perfect linear relationship. All points lie exactly on a straight line, so every change in X comes with a perfectly predictable change in Y .
- **$r = 0$** : no **linear** trend. A straight line does not help to predict Y .
- **Values in between** show how closely the points follow a straight line: the closer to -1 or $+1$, the stronger.
- The sign shows only the direction: $r = -0.8$ is exactly as strong as $r = +0.8$.

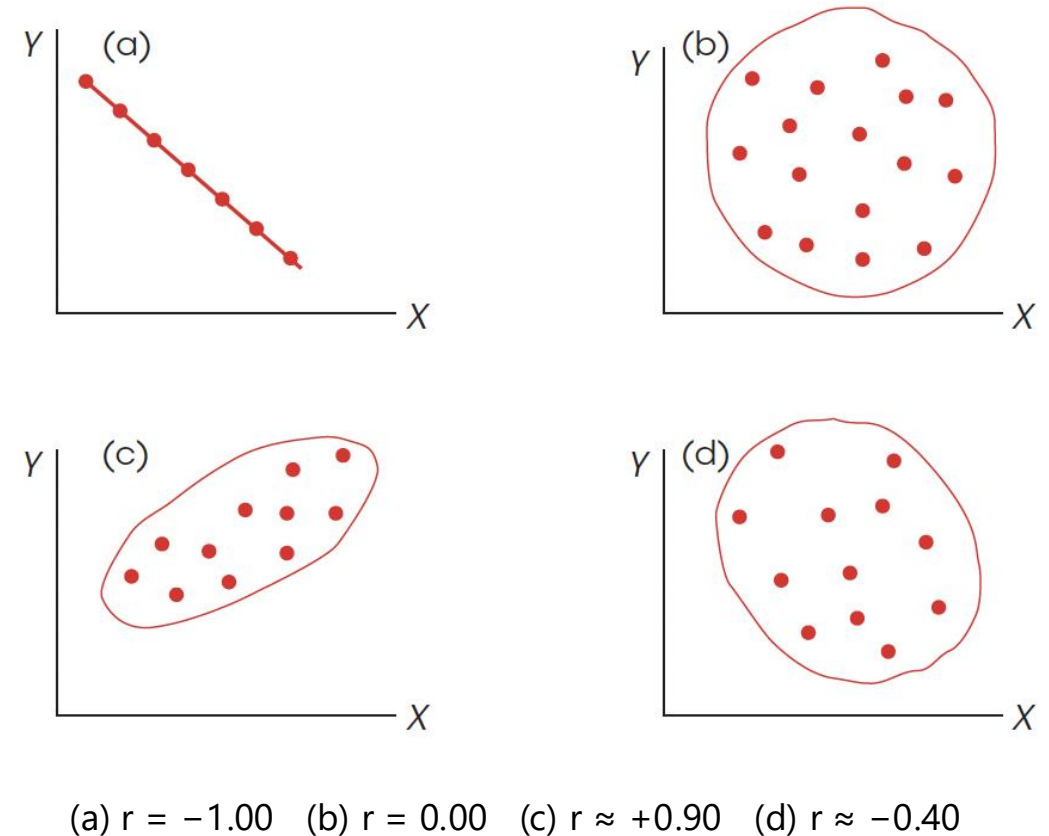
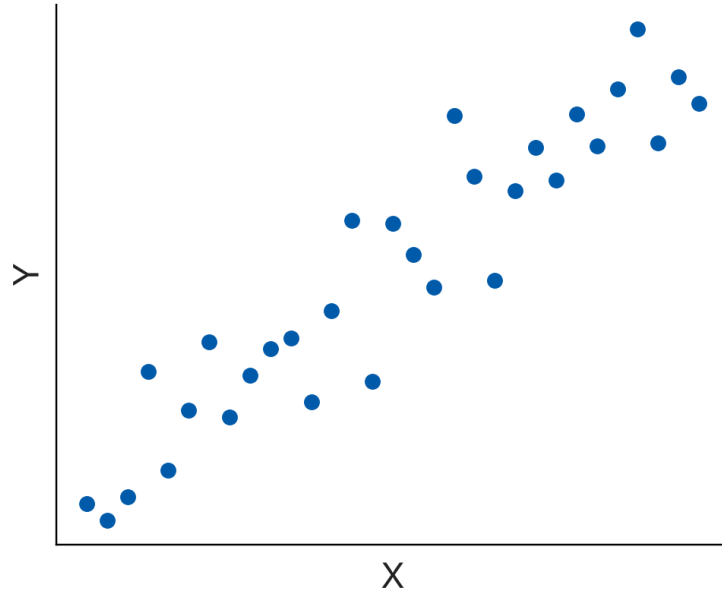


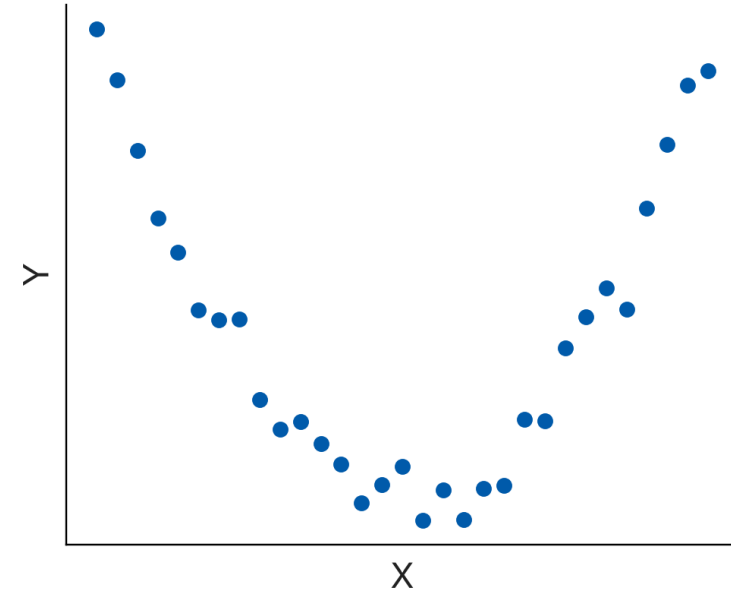
Figure: Gravetter & Wallnau, Statistics for the Behavioral Sciences

Form: r only sees straight lines

Linear relationship: $r = 0.94$



Curved relationship: $r = -0.05 \approx 0$



In both plots the relationship is strong: knowing X , we can predict Y quite well. But Pearson's r measures only how well a **straight line** fits the points. For the U-shaped relationship $r \approx 0$, although X and Y are clearly related.

Demographic example: mortality by age is U-shaped — high for infants, lowest in childhood, rising again in old age.

Always look at the scatter plot before trusting r .

Correlation coefficients

Coefficient $ r $	Usual description
from 0 to 0.3	very weak / small
from 0.3 to 0.5	weak
from 0.5 to 0.7	moderate
from 0.7 to 0.9	strong
from 0.9 to 1	very strong ($r = 1$: perfect linear relationship)

- For negative coefficients it is the same with a minus sign: $r = -0.6$ is a moderate negative correlation.
- These labels are **conventions**, and they differ between fields. In the social sciences relationships are rarely strong, so r around 0.3 can already be important (Cohen's guidelines: 0.1 small, 0.3 medium, 0.5 large).
- A correlation in a **small sample** can appear by pure chance: $r = 0.5$ from 8 people is much less convincing than $r = 0.3$ from 1 000 people. Whether a correlation is "real" is a question for **hypothesis testing** (next part).

Learning check

- 1.** A negative value for a correlation indicates _____.
 - a.** a much stronger relationship than if the correlation were positive
 - b.** a much weaker relationship than if the correlation were positive
 - c.** increases in X tend to be accompanied by increases in Y
 - d.** increases in X tend to be accompanied by decreases in Y

- 2.** Which of the following is the correct order, from strongest and most consistent to weakest and least consistent, for the following correlations?
 - a.** -1.00 , 0.85 , -0.43 , 0.02
 - b.** 0.85 , 0.02 , -0.43 , -1.00
 - c.** -1.00 , -0.43 , 0.02 , 0.85
 - d.** 0.85 , -0.43 , 0.02 , -1.00

Learning check: estimate the correlation

For each scatter plot, estimate the correlation: close to 0, moderate positive, or perfect positive?

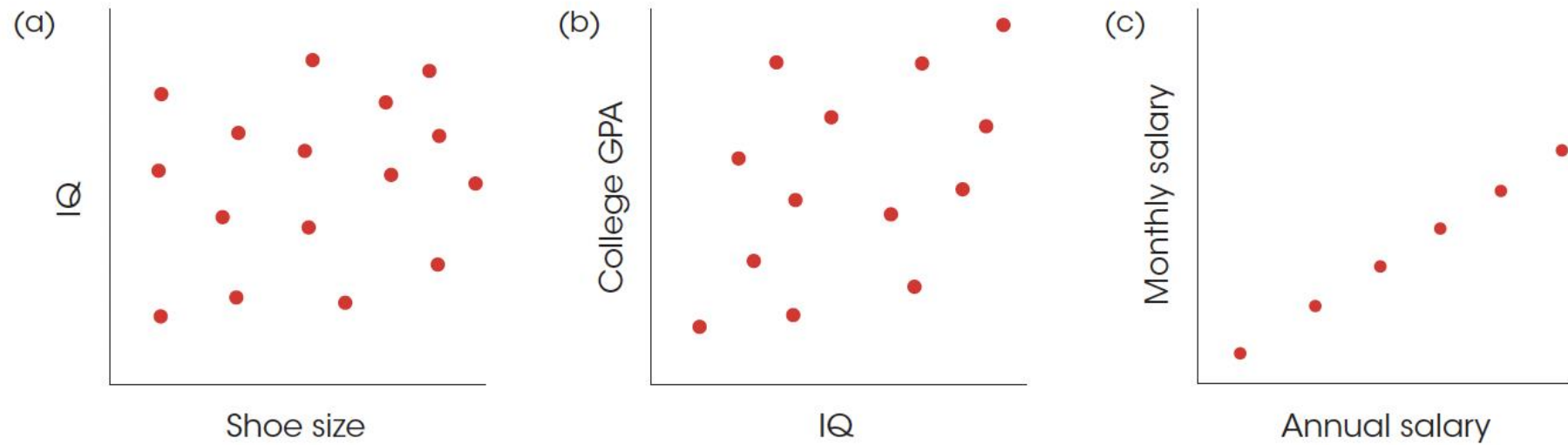


Figure: Gravetter & Wallnau, Statistics for the Behavioral Sciences

The Pearson correlation & formula

The **Pearson correlation** r measures the degree and the direction of the **linear** relationship between two variables.

Idea: r compares how much X and Y **vary together** with how much they **vary separately**.

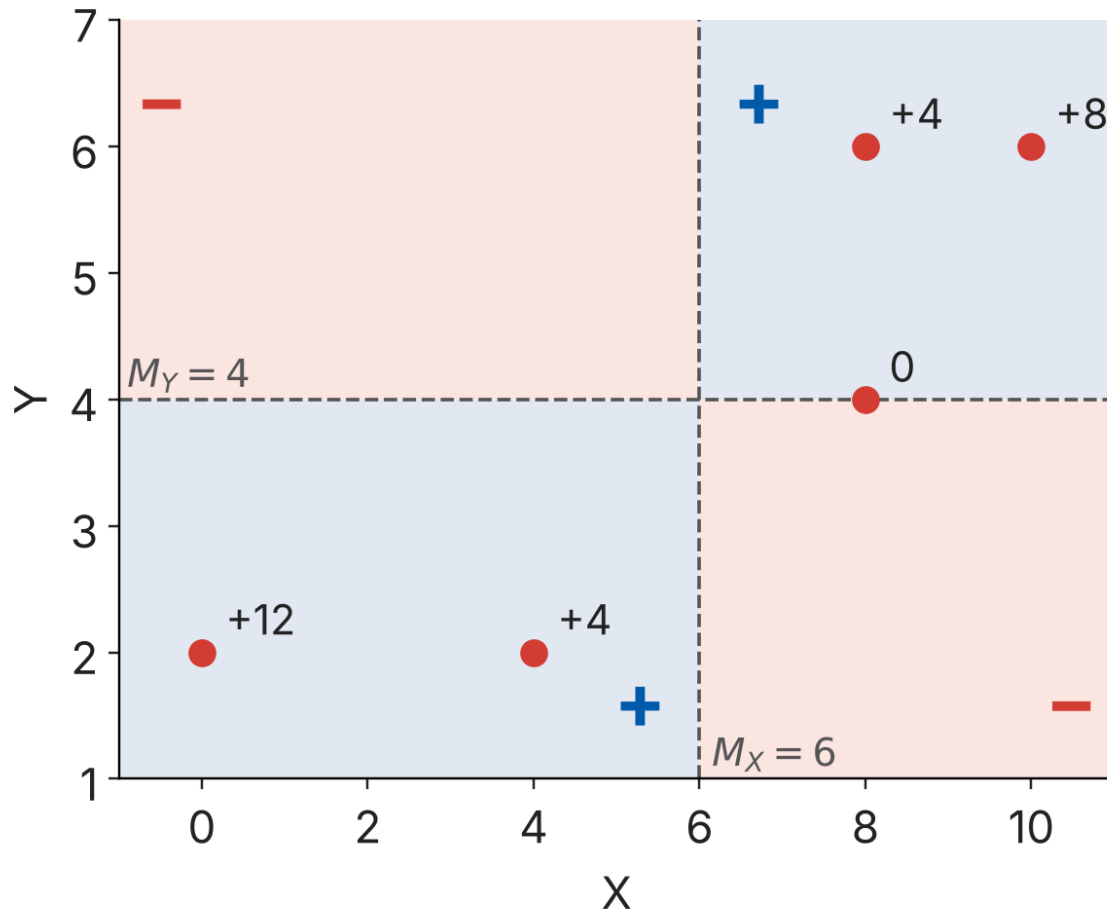
$$r = \frac{SP}{\sqrt{SS_X \cdot SS_Y}}$$

where

- $SP = \sum(X - M_X)(Y - M_Y)$ is the **sum of products** of deviations: how X and Y vary together;
- $SS_X = \sum(X - M_X)^2$ and $SS_Y = \sum(Y - M_Y)^2$ are the **sums of squared deviations**: how much X and Y vary separately;
- M_X and M_Y are the sample means of X and Y , and $\sum$ means "add up over all individuals".

Dividing by $\sqrt{SS_X \cdot SS_Y}$ removes the units of measurement, so r is always between -1 and $+1$.

Why the sum of products?



Draw lines at the means M_X and M_Y . They split the plot into four quarters.

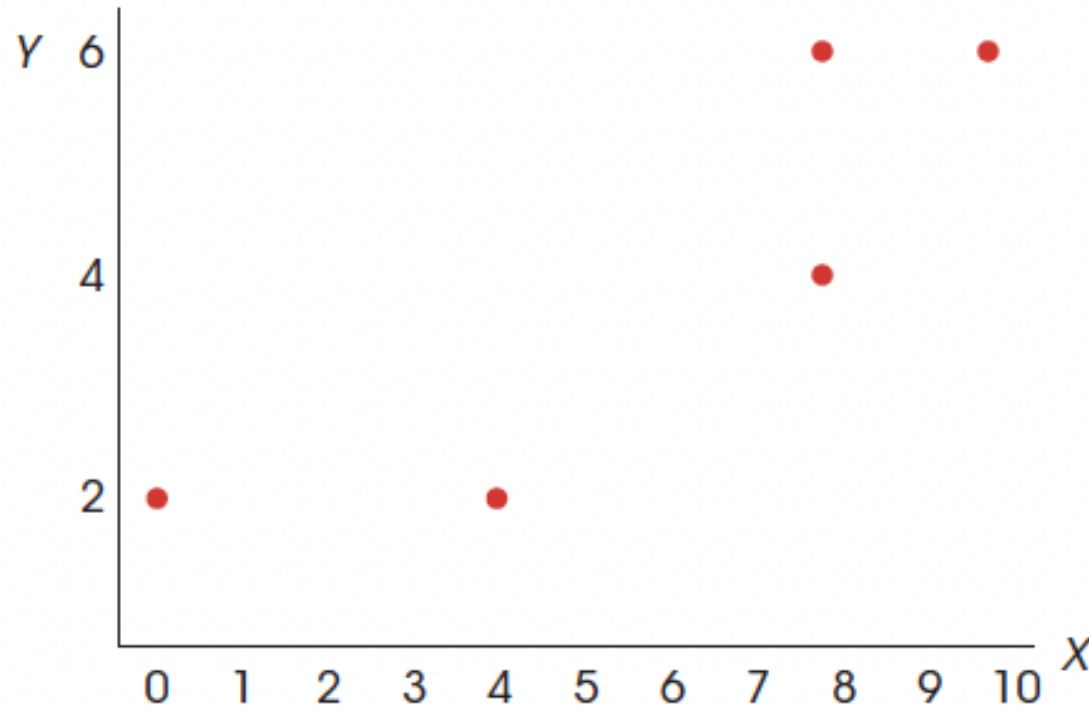
- Point **above both** means: $X - M_X > 0$ and $Y - M_Y > 0$, so the product is **positive**.
- Point **below both** means: both deviations are negative, and the product is again **positive**.
- One deviation positive, the other negative: the product is **negative**.

SP adds up all the products:

- most points in the "+" quarters $\rightarrow SP > 0$, positive relationship;
- most points in the "-" quarters $\rightarrow SP < 0$, negative relationship;
- points everywhere $\rightarrow$ products cancel out, $SP \approx 0$.

Calculating the Pearson correlation

Scores	
<i>X</i>	<i>Y</i>
0	2
10	6
4	2
8	4
8	6



Example: $n = 5$ individuals. Before computing anything, look at the scatter plot: the points tend to go up, so we expect a **positive** correlation.

Calculating the Pearson correlation

Scores		Deviations		Squared Deviations		Products
X	Y	$X - M_X$	$Y - M_Y$	$(X - M_X)^2$	$(Y - M_Y)^2$	$(X - M_X)(Y - M_Y)$
0	2	-6	-2	36	4	+12
10	6	+4	+2	16	4	+8
4	2	-2	-2	4	4	+4
8	4	+2	0	4	0	0
8	6	+2	+2	4	4	+4
				$SS_X = 64$	$SS_Y = 16$	$SP = +28$

$$r = \frac{SP}{\sqrt{SS_X \cdot SS_Y}} = \frac{28}{\sqrt{64 \cdot 16}} = \frac{28}{32} = +0.875$$

$r = 0.875$: a strong positive linear relationship, as the scatter plot suggested.

Learning check

1. What is the sum of products (SP) for the following data?

a. 6

b. -5

c. 43

d. None of the other 3 choices is correct.

X	Y
2	4
5	2
3	5
2	5

2. A set of $n = 5$ pairs of X and Y values has $SS_X = 5$, $SS_Y = 20$ and $SP = 8$. For these data, the Pearson correlation is _____.

a. $r = \frac{8}{100} = 0.08$

b. $r = \frac{8}{10} = 0.80$

c. $r = \frac{8}{25} = 0.32$

d. $r = \frac{8}{20} = 0.40$

When do we use correlation?

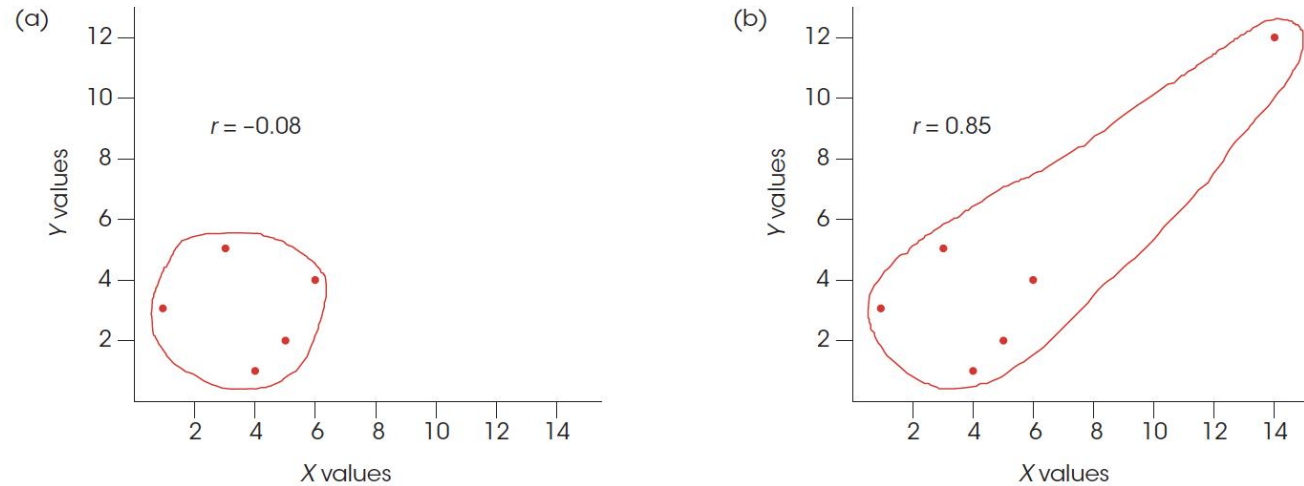
Prediction

If two variables are known to be related in some systematic way, one of them can be used to make predictions about the other. For example, school exam scores and university grades are correlated: students who do well at school exams tend to do well at university, so universities use exam scores for admission.

Interpreting correlation: **Cautions**

1. Correlation simply describes a relationship between two variables. It does not explain **why** they are related. A correlation should not and cannot be interpreted as proof of a **cause-and-effect** relationship.
2. The value of a correlation can be affected greatly by the **range of scores** in the data. For example, among students admitted with very high exam scores only, the correlation between scores and grades is much weaker than among all students.
3. One or two extreme data points, called **outliers**, can have a dramatic effect on the value of a correlation.

Correlation and outliers



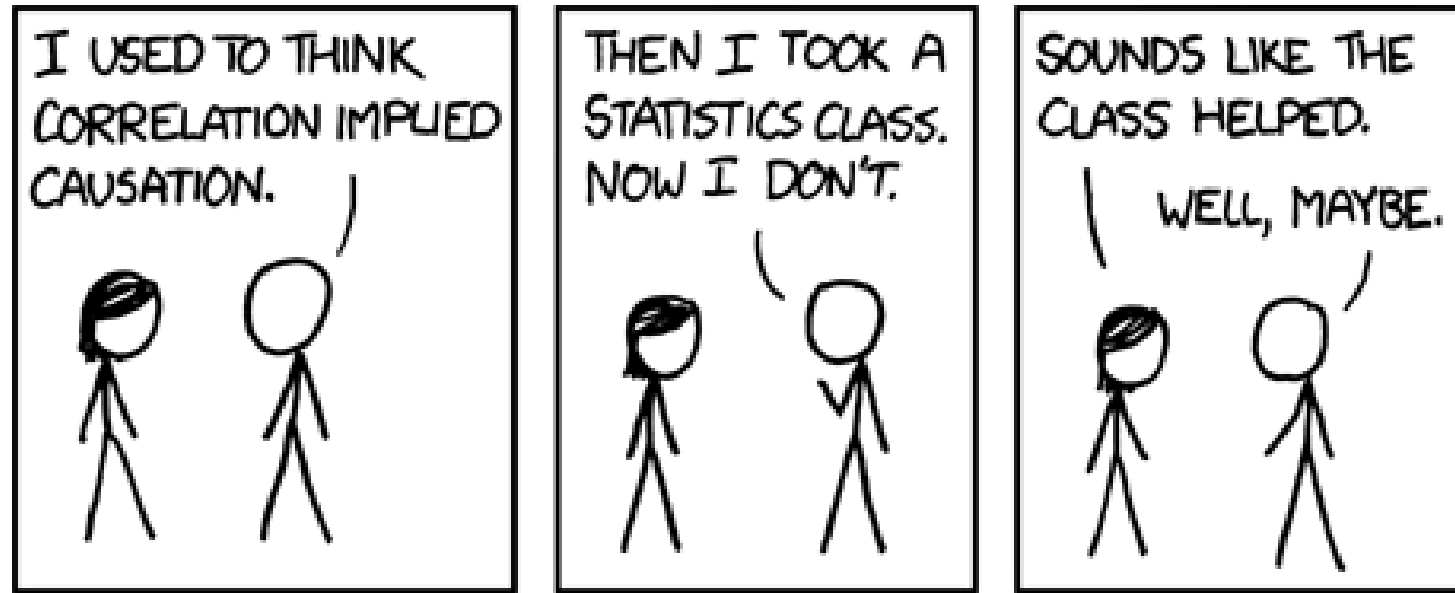
One or two extreme points (**outliers**) can change the correlation dramatically.

- Left: 5 points without any clear trend, $r = -0.08$.
- Right: the same 5 points plus **one** extreme point at (14, 12). Now $r = 0.85$.

A single individual created a "strong" correlation. The other points still show no relationship.

Always check the scatter plot for outliers.

Correlation does not imply causation



Although there may be a causal relationship, the simple existence of a correlation does not prove it.

Example: countries where people own more TV sets have higher life expectancy. TV sets do not make people live longer: both are driven by a third variable — **wealth**.

To establish cause and effect, we need a true experiment, where the researcher manipulates one variable and controls the others, or a careful study design that rules out third variables.

Correlation does not imply causation

Correlation means there is a statistical relationship or pattern between two variables. When one changes, the other tends to change too.

Causation means that one event *directly causes* the other to happen.

The Golden Rule: Just because two things happen together, it does not mean one caused the other. Usually, there is a **Third Variable (Confounding Variable)** driving both, or it is a pure coincidence.

- **Ice Cream Sales and Shark Attacks** *The Correlation:* As ice cream sales increase, the number of shark attacks also rises.

- *The Flawed Conclusion:* Eating ice cream makes you attractive to sharks (or sharks make people buy ice cream).

- **Firefighters and Fire Damage** *The Correlation:* The more firefighters sent to a fire, the more damage the house suffers.

- *The Flawed Conclusion:* Firefighters are causing the destruction.

- *The Reality (Third Variable):* **The size of the fire.** Larger, more dangerous fires require more firefighters and naturally cause more damage.

- **Nicholas Cage Movies vs. Drownings** *The Fact:* The number of people who drowned by falling into a pool correlates strongly with the number of films Nicolas Cage appeared in that year.

Hypothesis Testing



Hypothesis test

In a sample we can see a difference or a correlation. But is it real for the whole **population**, or did we just get a lucky sample? A **hypothesis test** is a statistical method that uses sample data to evaluate a hypothesis about a population.

➤ Step 1. Posing a hypothesis

First, **we state a hypothesis about a population**. Usually, the hypothesis concerns the value of a population parameter, for example the population mean μ .

Example: HSE students on average drink more coffee during the examination week than during a regular week.

➤ Step 2. Predicting what the sample should look like

Before we select a sample, we use the hypothesis of "**no effect**" to predict what the sample should look like.

If the examination week changes nothing, the average change in coffee consumption in our sample should be close to 0 — but not exactly 0.

Remember: the sample should be similar to the population, but you always expect a certain amount of **random error**!

Hypothesis test

➤ Step 3. Selecting a sample

Next, we obtain a random sample from the population.

Example: we select $n = 100$ HSE students and measure the coffee consumption of each student in a regular week and in the examination week.

➤ Step 4. Making a conclusion

Finally, we compare the sample data with the prediction made from the "no effect" hypothesis.

If the sample is **consistent** with "no effect", we have no evidence of an effect: we **fail to reject** the "no effect" hypothesis. (This does not prove that there is no effect.)

If the sample is **very unlikely** under "no effect", we **reject** it and conclude that there is an effect.

Our data: one number per student

Student	Regular week, ml	Exam week, ml	Difference D = exam – regular
A	700	900	+200
B	0	0	0
C	1 500	1 400	–100
D	300	650	+350
Mean	625	737.5	+112.5

- Bonus: every student is compared with himself or herself, so the big differences between heavy and light coffee drinkers cancel out.
- If the two weeks were measured on **different** students, we could not compute D. Then a two-sample test is needed.

Each student is measured **twice**, so for each student we compute **one number**: the difference D.

The mean of the differences equals the difference of the means:

$$112.5 = 737.5 - 625$$

This is always true, not only in this example.

So the question "is coffee consumption in the exam week different?" becomes a question about **one mean**: is the mean difference equal to 0?

Posing a hypothesis

The **null hypothesis** H_0 states that there is no change, no effect, no difference — nothing happened, hence the name null.

$$H_0: \mu_D = 0 \quad (\text{on average, the exam week changes nothing})$$

The **alternative hypothesis** H_1 states that there is a change, a difference, or a relationship in the population.

$$H_1: \mu_D \neq 0 \quad (\text{on average, coffee consumption changes})$$

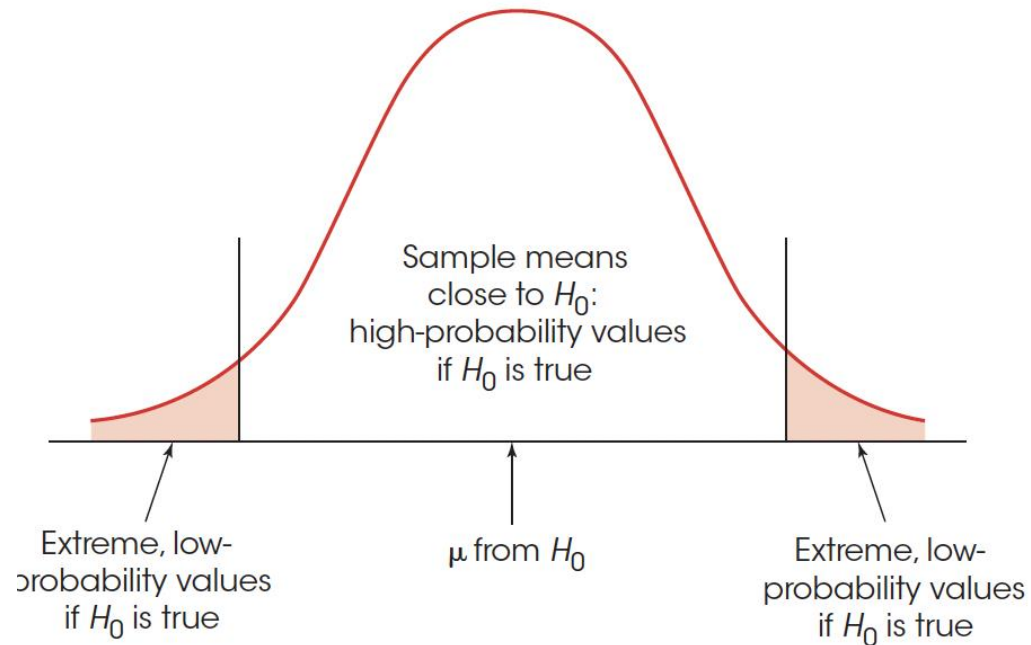
- μ_D is the mean difference in the whole **population** of students. It is unknown. We estimate it by the mean difference in our **sample**, M_D .
- Why do we test H_0 and not H_1 ? H_0 gives one exact value (0), so we can predict what samples should look like. H_1 ("some change") does not give an exact value.
- $H_1: \mu_D \neq 0$ is **two-sided**: we look for a change in either direction. If only an increase matters, a one-sided $H_1: \mu_D > 0$ is possible; we use the standard two-sided version.

Set criteria for decision

To make a decision, we use the null hypothesis to predict which sample means are **likely** and which are **unlikely**.

Sample means close to the value from H_0 are likely if H_0 is true. Sample means very far from it are very unlikely if H_0 is true.

To find exactly where "close" ends and "very far" begins, we need to know all the sample means we could get if H_0 is true: the **distribution of sample means**.



Distribution of sample means

Sample	First	Second	Sample mean M
1	2	2	2
2	2	4	3
3	2	6	4
4	2	8	5
5	4	2	3
6	4	4	4
7	4	6	5
8	4	8	6
9	6	2	4
10	6	4	5
11	6	6	6
12	6	8	7
13	8	2	5
14	8	4	6
15	8	6	7
16	8	8	8

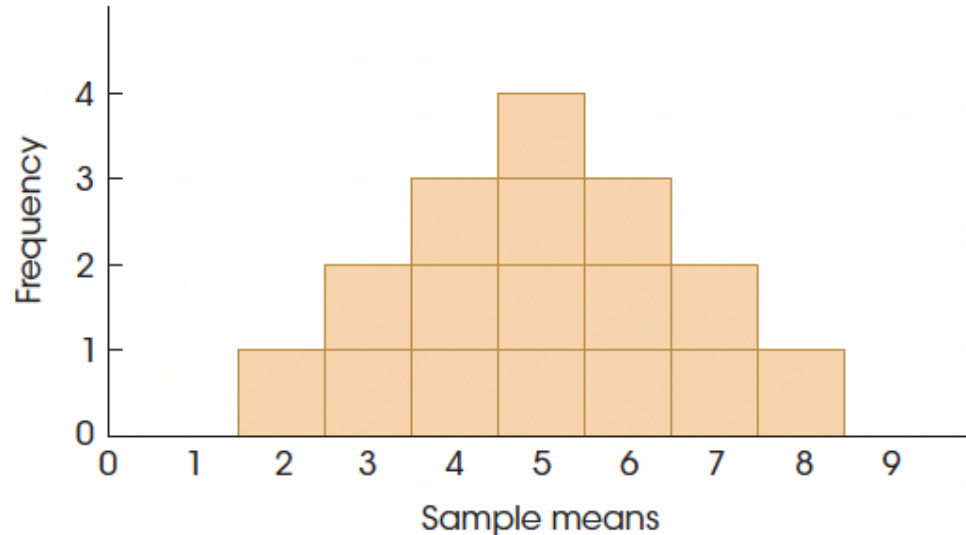
Consider a population that consists of only 4 scores: **2, 4, 6, 8**. Its mean is $\mu = 5$.

We use this population to build the distribution of sample means for samples of $n = 2$ scores. The first and the second score can be the same, so there are $4 \times 4 = \mathbf{16}$ different samples.

Next, we compute the mean M of each of the 16 samples.

Thus, we list **all the means we could get** if we randomly obtained ONE sample from our population.

Distribution of sample means



Looking at the distribution of sample means, we can ask about the **probability** of obtaining a sample with a particular mean.

What is the probability of a sample mean greater than 7?

$$p(M > 7) = 1/16 \approx 0.06$$

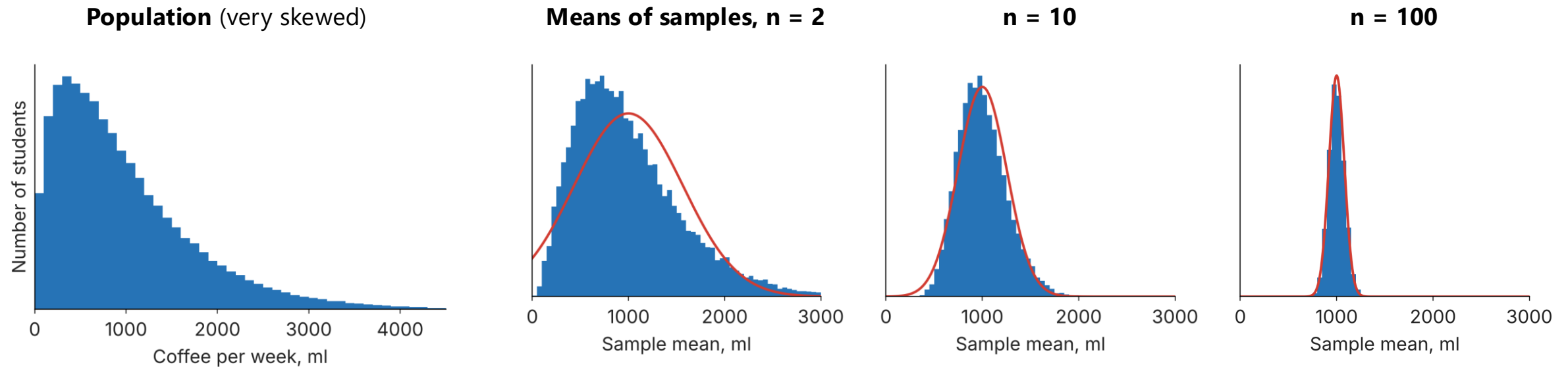
Only one sample (8, 8) has such a mean: **very unlikely!**

On the other hand, a sample mean equal or close to the population mean is **very likely**:

$$p(4 \leq M \leq 6) = 10/16 \approx 0.63$$

The distribution of sample means is centered at the population mean $\mu = 5$.

Bigger samples: the Central Limit Theorem



Coffee per week of all students: $\mu = 1\,000$ ml, $\sigma = 800$ ml (σ is the standard deviation). Most students drink little, a few drink a lot.

Central Limit Theorem (CLT): whatever the shape of the population, for a large enough sample (rule of thumb: $n \geq 30$) the distribution of the sample mean M

1. is centered at the population mean μ ;
2. has standard deviation $\sigma / \sqrt{n}$, called the **standard error** (SE): it shrinks as n grows ($n = 100$: $SE = 800 / \sqrt{100} = 80$ ml);
3. is approximately **normal** (red curve). For $n = 2$ the normal curve does not fit yet; for $n = 100$ it fits almost perfectly.

The Central Limit Theorem

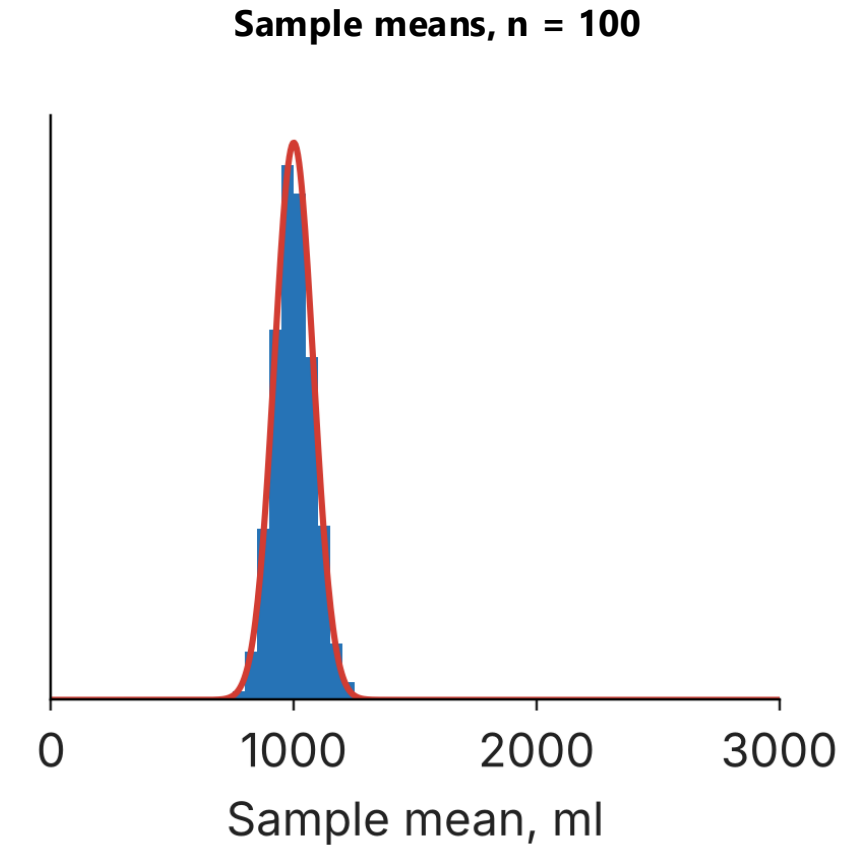
Take n independent observations from a population with mean μ and standard deviation σ . The sample mean M is random: another sample would give another M .

For any n (exactly, derived on the next slide):

- M is centered at the population mean μ ;
- its standard deviation is $\sigma / \sqrt{n}$, the **standard error** (SE).

Central Limit Theorem: as $n \rightarrow \infty$, the distribution of M becomes **normal**, whatever the shape of the population. More precisely, the distribution of $z = (M - \mu) / (\sigma / \sqrt{n})$ tends to the standard normal distribution. In practice $n \geq 30$ is usually enough (more for very skewed data).

The CLT says nothing about repeating the study. It describes all the samples of size n we **could** have obtained. In a real study we have **one** sample, and the CLT tells us whether its mean is typical or rare under H_0 .



Why the spread shrinks: standard error

Each student deviates from μ by a typical amount σ . It is easier to add up spread in squared units, the **variance** σ^2 .

1. For independent students, the variances of a sum **add up**:

$$\text{Var}(X_1 + \dots + X_n) = n \cdot \sigma^2$$

2. The mean M is the sum divided by n . Dividing by n divides the variance by n^2 :

$$\text{Var}(M) = \frac{n \cdot \sigma^2}{n^2} = \frac{\sigma^2}{n}$$

3. The square root brings us back to the original units (ml):

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

z-score: how far from H_0 ?

By the CLT, if H_0 is true, the sample mean M is approximately normal, centered at the value from H_0 (call it μ_0 ; in our example $\mu_0 = 0$), with spread SE .

We measure the distance between our sample mean and μ_0 **in standard errors**:

$$z = \frac{M - \mu_0}{SE} = \frac{M - \mu_0}{\sigma / \sqrt{n}}$$

- $z = 1$ means "one standard error away from what H_0 predicts"; $z = 2.5$ means "two and a half standard errors away".
- If H_0 is true, z follows the **standard normal distribution** (mean 0, standard deviation 1), whatever the units of the data.
- For the standard normal distribution we know exactly how often each distance happens: **95%** of all z -values fall between **-1.96 and +1.96**.

The Alpha Level

To find the boundaries that separate the high-probability samples from the low-probability samples, we must define exactly what is meant by "low" probability and "high" probability.

This is done by selecting a specific probability value, called the **level of significance**, or the **alpha level** α .

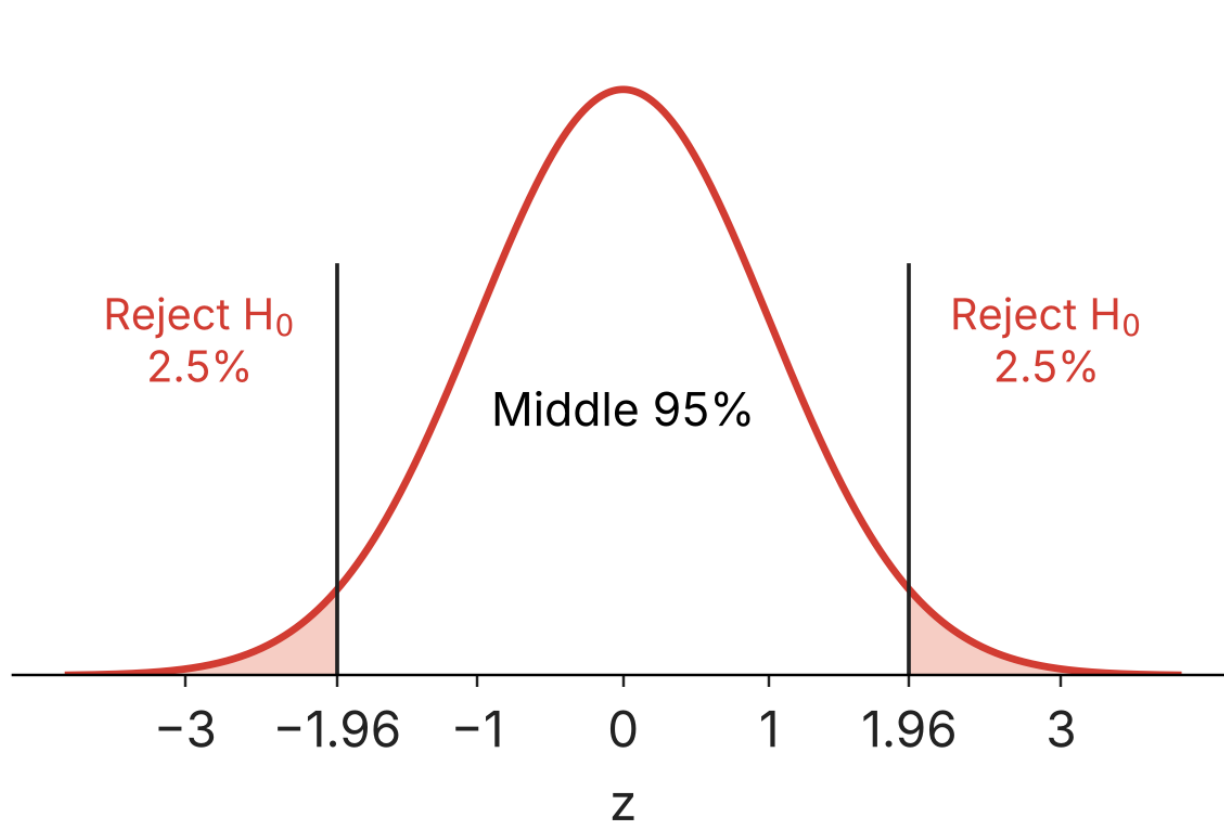
The alpha value is a small probability that is used to identify the low-probability samples. By convention, commonly used alpha levels are $\alpha = 0.05$ (5%), $\alpha = 0.01$ (1%), and $\alpha = 0.001$ (0.1%).

For example, with $\alpha = 0.05$, we separate the most unlikely 5% of the sample means (the extreme values) from the most likely 95% (the central values).

α is also the probability of a **false alarm** (Type I error): rejecting H_0 when it is actually true. With $\alpha = 0.05$ we accept that in 5% of studies where nothing happened, we will wrongly "find" an effect.

Critical Region

The **critical region** is composed of the extreme values that are very unlikely (as defined by the alpha level) if the null hypothesis is true. If our statistic falls in the critical region, H_0 is rejected.



With $\alpha = 0.05$ and a two-sided H_1 , we cut 2.5% from each tail of the standard normal distribution.

The boundaries are $z = -1.96$ and $z = +1.96$.

Decision rule:

- $|z| > 1.96 \rightarrow \text{reject } H_0$;
- $|z| \leq 1.96 \rightarrow \text{fail to reject } H_0$.

Coffee example: the z-test step by step

1. **Hypotheses and alpha level.** $H_0: \mu_D = 0$, $H_1: \mu_D \neq 0$, $\alpha = 0.05$.
2. **Critical region.** Reject H_0 if $|z| > 1.96$.
3. **Compute the statistic.** Our data: $n = 100$ students, mean difference $M_D = 100$ ml, standard deviation of the differences $s = 400$ ml. We do not know σ , so we use the sample value s instead.

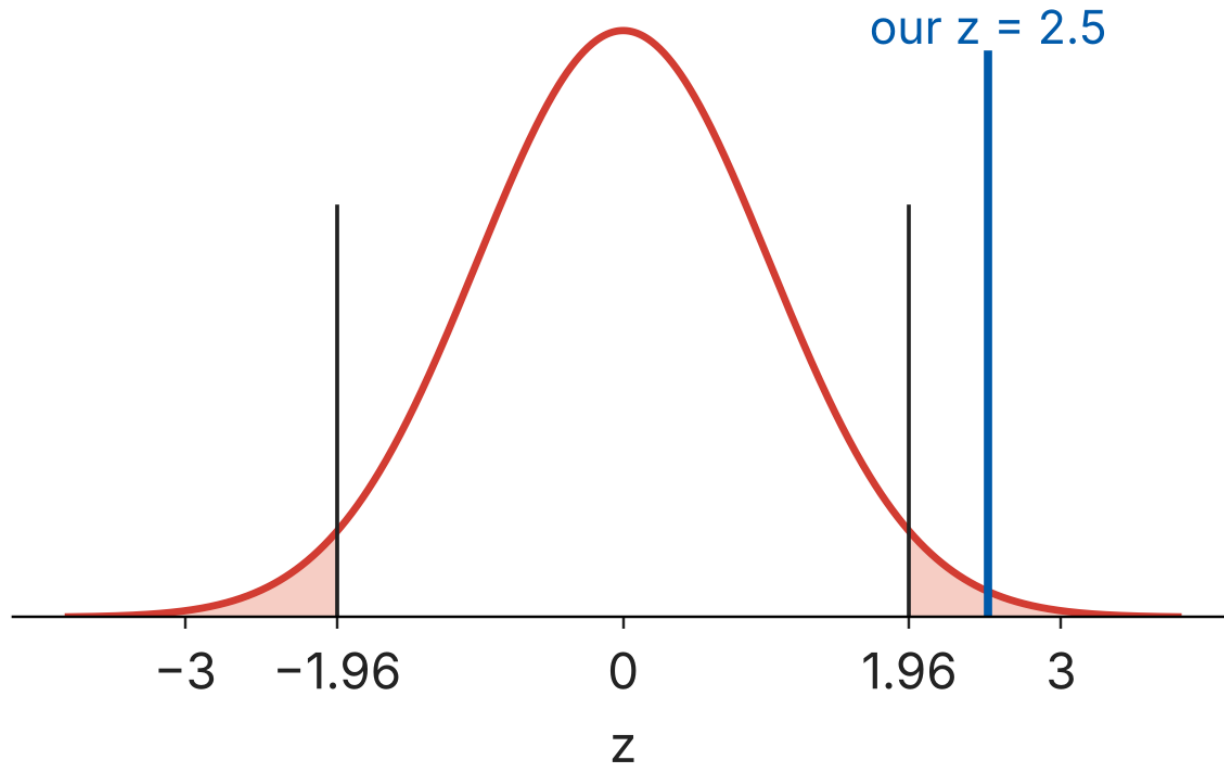
$$SE = \frac{s}{\sqrt{n}} = \frac{400}{\sqrt{100}} = \frac{400}{10} = 40 \text{ ml}$$

$$z = \frac{M_D - 0}{SE} = \frac{100}{40} = 2.5$$

4. **Decision.** $z = 2.5 > 1.96$: our sample mean is 2.5 standard errors away from 0, inside the critical region. We **reject H_0** .

Conclusion: HSE students drink more coffee during the exam week (on average +100 ml per week).

Where is our z?



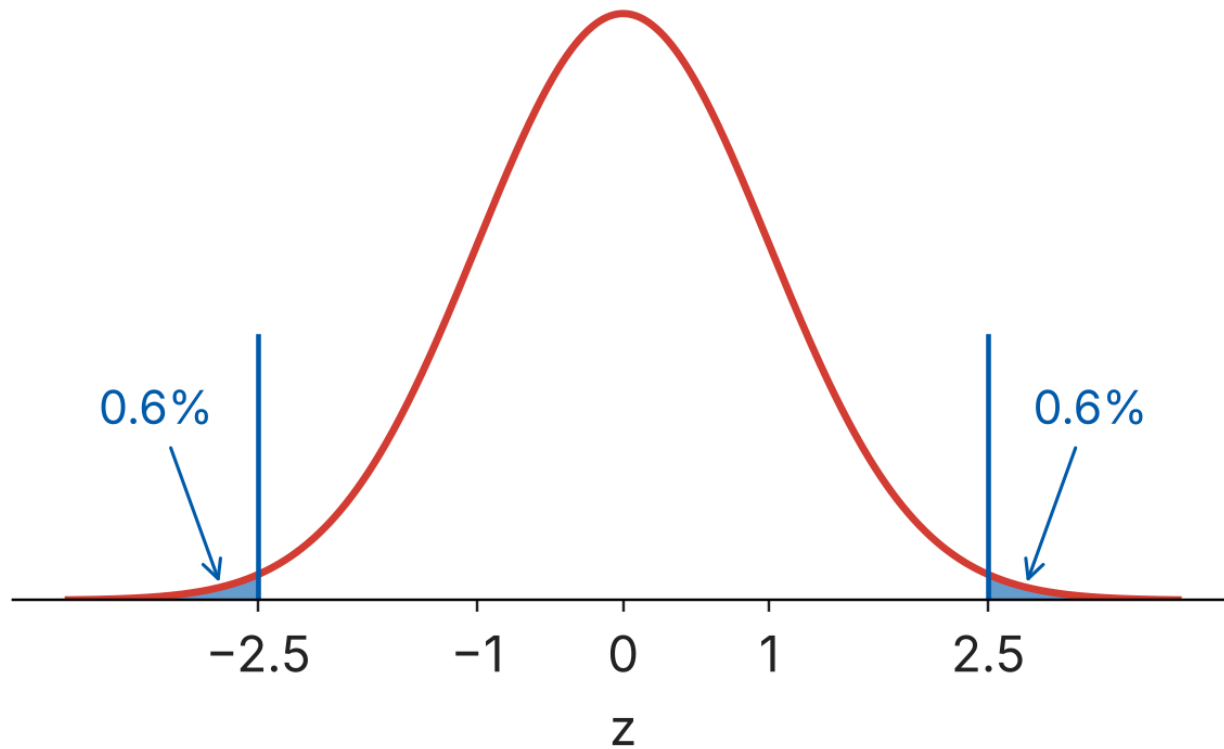
Our $z = 2.5$ falls into the critical region.

If H_0 were true (no change at all), a sample mean this far from 0 would be very unlikely: it belongs to the most extreme 5% of all possible samples.

So we reject H_0 : the data are not consistent with "no change".

"Reject H_0 " is a decision with a known risk of error ($\alpha = 5\%$), not a proof.

The p-value



Software (for example, Python) usually reports a **p-value** instead of a critical region.

p-value = the probability of getting a result at least as extreme as ours **if H_0 were true**.

For $z = 2.5$: the area beyond -2.5 and beyond $+2.5$ is $0.6\% + 0.6\%$, so $p \approx 0.012$.

Decision rule: $p < \alpha \rightarrow$ reject H_0 . Here $0.012 < 0.05$, so we reject H_0 : the same decision as $|z| > 1.96$. Both say "our result is among the extreme 5%".

Caution: the p-value is **not** the probability that H_0 is true.

Remarks and cautions

1. **σ is usually unknown.** We used the sample standard deviation s instead. Strictly speaking, this gives a **t-test**: its critical value is slightly larger (for $n = 100$: 1.98 instead of 1.96). For large samples the difference is negligible; for small samples use the t-test.
2. **Paired vs independent samples.** We could compute a difference for each student because the same students were measured twice. Two different groups of students need a two-sample test.
3. **Statistically significant $\neq$ important.** With a huge sample even a tiny effect (say +5 ml per week) becomes significant. Always look at the size of the effect: here +100 ml per week, about half a cup.
4. **Fail to reject $\neq$ prove H_0 .** A non-significant result means "not enough evidence", not "no effect".

Thank you!